What Do Neural Predictors Add to Budget Pacing?

Spend Forecasts, Budget Delivery and Hedging on Criteo Campaign Logs

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Abstract

An advertising platform can pace a daily budget by forecasting the cost of the remaining impressions and choosing what share to keep. Does a neural forecast improve this decision? We compare five neural predictors with seasonal-naive and LightGBM forecasts on 414 Criteo campaigns, using a fixed impression-replay rule at three budget levels. The target is hourly spend in Criteo's transformed cost units, not impression counts.

All five neural predictors capture more clicks than the seasonal-naive plan: 0.98–2.14 additional percentage points of the hindsight-reference click count. They spend 1.2–2.1% more, with little change in clicks per unit of spend. Multiplying the naive forecast by a validation-chosen scale recovers 36–94% of TFT's gain. In a separate experiment that swaps daily totals and hourly profiles, the TFT total accounts for 68–82% of the improvement. An exploratory comparison at equal aggregate spend leaves all five models 0.26–1.30 points above the naive scaling curve; this does not identify which part of the forecast causes the remaining gain. For the four forecasters in the original study, planning with 90th-percentile hourly forecasts loses 7.7–10.6 points relative to median forecasts. Calibration recovers only part of that loss.

These findings cover 5 test days in a simulator that barely penalizes early budget exhaustion. The intervals exclude training-seed uncertainty. A small iPinYou comparison does not reproduce the policy ordering and does not test neural predictors. The evidence supports a gain on Criteo, not a general advantage across advertising markets.

Keywords

budget pacing, throttling, real-time bidding, probabilistic forecasting, neural predictors, empirical study

1 Introduction

An advertiser's daily budget may run out before the day ends, or remain partly unspent. Budget pacing controls how quickly the platform spends it [1, 14]. One approach is *probabilistic throttling*: the platform participates in each eligible auction with probability α_t, which it updates during the day [1, 2, 20].

A reactive controller adjusts this probability using past spend. A proportional-integral-derivative (PID) controller, for example, tracks the gap between actual spend and a target schedule. A forecast-based rule instead estimates how much the remaining impressions would cost. It divides the remaining budget by that estimate to choose the participation rate. Forecasting too much spend lowers participation and may leave budget unused.

We ask whether neural predictors improve this rule, and what information they add. A spend forecast provides both a daily total and an hourly profile: the share of the total assigned to each hour. A better total could help even if the hourly profile adds little. We also test whether using an upper quantile, rather than the median hourly forecast, protects against early exhaustion at an acceptable cost in clicks.

The main study uses the public Criteo Attribution dataset [8]. We compare the Temporal Fusion Transformer (TFT), NHITS, PatchTST, a long short-term memory (LSTM) model and TiDE with seasonal naive and LightGBM. The initial study tested TFT and NHITS; the other three neural models were added later. TFT remains the reference for the experiments that examine totals, profiles and feedback.

The paper contributes an hourly spend-forecasting and replay pipeline, a comparison with forecast-free policies, and experiments that separate the benefits of spending more from changes in clicks per unit of spend. We then test forecast scaling, total–profile swaps, perfect remaining-spend knowledge, and conservative planning. A generated audience environment and a small iPinYou study examine how much of the pattern carries beyond Criteo.

These are empirical comparisons of established models and pacing rules, not a new architecture or a theoretical guarantee. The main results use 5 test days. Follow-up experiments were designed after those results were known; their timing and decision rules are disclosed in Section 5 and the accompanying protocol. All reported measurements come from saved results files. Code, protocol and results files will be released.

2 Related Work

Pacing rules. Lee et al. [14] study bidding with smooth budget delivery. Agarwal et al. [1] use supply patterns to pace campaigns at LinkedIn. Smart pacing [20] learns participation rates for different segments. Dai et al. [6] control probabilistic throttling with dual multipliers and report online tests on Taobao. Balseiro et al. [2] compare budget-management strategies in equilibrium, while Zhang et al. [22] study optimal bidding under budget constraints.

Zhang et al. [21] use feedback to adjust bids toward winning-rate and cost-per-click targets. Our PID baseline adjusts participation instead of bid prices. Chen's practical guide [5] covers throttling, PID, model-predictive control and dual online-gradient methods. It notes that a PID controller cannot repair a misspecified target merely by tracking it, and discusses updating the remaining-supply forecast.

Forecasts and decisions. Shekhar and Howard [18] calibrate uncertainty according to its effect on pacing decisions. Their experiments use replays calibrated from Criteo Uplift and KuaiRand, rather than the Criteo Attribution replay used here. They find that the lowest forecast error need not give the smallest pacing-specific uncertainty radius. We study a different question: how much changing the forecast improves clicks under a fixed throttling rule.

TFT [15] combines recurrent encoding, variable selection and attention. NHITS [4] uses multi-rate sampling and hierarchical interpolation. We use their `neuralforecast` implementations, along

with the library's PatchTST [16], LSTM [9] and TiDE [7] models [17]. LightGBM [12] provides a non-neural learned baseline. This study evaluates these models as forecasting tools; it does not propose changes to their architectures.

3 Pacing Setup

What is forecast. The target is the sum of logged impression costs in each campaign-hour. We call this *spend demand*: the cost of keeping all impressions in that part of the log, before applying our throttle. It is neither an impression count nor a forecast of spend after pacing. Criteo transforms the costs, so spend and budget are measured in transformed cost units, not currency.

Episodes and budgets. One episode is a campaign-day of 24 hours. The budget is $B = \rho \bar{D}_c$, where $\bar{D}_c$ is campaign c's mean daily *training* spend. We use $\rho \in \{0.3, 0.5, 0.7\}$, called the tight, medium and loose budgets. Logged daily spend demand exceeds the budget in 94.3%, 86.5% and 70.8% of test episodes, respectively.

Replay. At the start of hour h, the policy sets a retention probability $\alpha_h \in [0, 1]$. Each logged impression is kept independently with that probability. Within the hour, kept impressions contribute their logged cost, click and attributed conversion in chronological order, stopping before the next kept impression would exceed the remaining budget. The same uniform random draws are used for each policy, campaign, hour and replay seed.

This is a replay of *won impressions*, not a simulation of all auctions. The original bidder's choices already shaped the log. We assume that keeping fewer impressions does not change their prices or outcomes. The log cannot verify this counterfactual, and the replay does not model bid-price changes or competitors' responses.

Policies. Let S_h be cumulative spend before hour h and $R_h = B - S_h$ the remaining budget.

- *No pacing*: keep every impression that fits the remaining budget ($\alpha_h = 1$).
- *Even pacing*: start at $\alpha_0 = \min(1, B/\bar{D}_c)$ and adjust the previous rate multiplicatively toward spending $R_h/(24-h)$ per remaining hour.
- *PID*: use the normalized spending gap $e_h = (Bh/24 - S_h)/B$ to set $\alpha_h = \alpha_0 + K_p e_h + K_i \sum_{j \le h} e_j + K_d(e_h - e_{h-1})$. The replay clips the rate to $[0, 1]$. Gains are grid-searched on validation days separately for each budget level.
- *Forecast throttle*: forecast hourly spend demand for the rest of the day and set

$$\alpha_h = \min\Big(1,\ R_h \Big/ \sum_{j=h}^{23} \hat{y}_j^{(q)}\Big), \qquad q \in \{0.5, 0.7, 0.9\}. \tag{1}$$

  Here $\hat{y}_j^{(q)}$ is an hourly quantile forecast issued at the start of hour h. The plan is updated every hour.
- *Hindsight constant*: use the fixed rate $\alpha = \min(1, B/D)$, where D is the day's logged spend demand. This reference supplies the denominator of our click ratio. It is not an upper bound: a time-varying policy can capture more clicks.

Using a higher q usually increases the denominator in Eq. (1), reducing participation. We call this *hedging* against early budget exhaustion. The denominator sums hourly quantiles; it is not generally the same quantile of total remaining spend demand.

Scaling and fixed plans. The scaled-naive policy multiplies the median-forecast denominator by s. A smaller s raises participation. One scale per budget level is selected on validation days from 25 values between 0.50 and 1.10. For total–profile swaps, we hold the day-start forecast fixed. Its sum supplies the daily total and its normalized hourly values supply the profile. We take these two components from either TFT or seasonal naive. An *oracle forecast* replaces the predicted total or remaining spend demand with its actual logged value; the throttling rule is unchanged.

Calibrated hedging. Instead of summing upper hourly quantiles, we multiply the median plan by a factor $f_q(h)$. The factor is the empirical q-quantile of actual remaining spend demand divided by the median plan, for that hour. Calibration uses all campaigns on earlier test days only. The comparison therefore covers the last 4 test days. A separate cross-fitted variant uses factors from other campaigns, including their same-day data; it is not a deployable causal forecast.

Feedback and dual-gradient control. We add a PID correction to the forecast rate, first using a spending schedule from the day-start forecast, then using the previous hour's updated plan. Feedback gains are tuned with seasonal naive and transferred unchanged to TFT and NHITS. We also test a reactive dual-gradient rule, a throttling version of dual-gradient bid pacing [3], $\alpha_h = \min(1, 1/(1+\mu_h))$. Each hour its nonnegative multiplier increases when spend exceeds an hourly target and decreases when spend falls short. Targets are either flat ($B/24$) or based on the campaign's training-hour profile. Step sizes are chosen on validation days.

4 Forecasting Models

Each forecast predicts the next 24 hours of campaign spend demand, using only information available before its issue hour. We issue forecasts every hour without refitting the models. All forecasters output quantiles $\{0.1, 0.5, 0.7, 0.9\}$; pacing uses the hours remaining in the current day.

- *Seasonal naive*: take empirical quantiles of spend in the same hour on the previous seven days. At $q = 0.5$, this is their median.
- *LightGBM*: fit one quantile-regression model per quantile, with forecast step as an input. Other inputs are spend lags, rolling statistics, calendar phase and campaign statistics computed on training hours. Spend is normalized by the campaign's training mean.
- *TFT* and *NHITS*: take 168 hours of past data. Both use multi-quantile loss and robust scaling. Inputs include spend, past impressions and clicks, known calendar phase and training-derived campaign statistics.
- *PatchTST*, *LSTM* and *TiDE*: add a patch-based transformer, a recurrent model and a dense encoder–decoder. They use the same training procedure, with architecture-specific settings fixed before their runs. LSTM and TiDE use the covariates above. PatchTST uses only past spend.

Learned models are fitted on training days and early-stopped using validation days. Each neural model has 3 training seeds; LightGBM has one run. Hyperparameters are listed in the protocol and code. The original comparison covers TFT, NHITS and the two non-neural forecasters. PatchTST, LSTM and TiDE were added under protocol entry 21 and evaluated for median-plan pacing. The hedging and feedback experiments were not repeated for these three additions.

Table 1: Original four-forecaster study. Click ratios pool episodes and seeds; bold marks the best non-reference value. Operational metrics are shown at $\rho = 0.5$.

	Clicks / reference			At $\rho = 0.5$			
Policy	ρ=0.3	ρ=0.5	ρ=0.7	Conv./ref.	BUR (%)	Early depl. (%)	PVI
Hindsight constant (ref.)	1.000	1.000	1.000	1.000	90.6	9.5	0.000
No pacing	0.985	0.979	0.988	1.018	94.4	80.1	0.000
Even pacing	0.993	0.960	0.942	0.966	90.2	6.4	0.314
PID (tuned)	1.009	0.990	0.988	1.004	92.2	31.9	0.577
Seasonal naive, q=0.5	1.007	0.989	0.983	0.988	91.3	6.4	0.083
LightGBM, q=0.5	1.019	0.997	0.992	0.998	92.4	5.0	0.094
NHITS, q=0.5	**1.028**	**1.007**	0.995	1.005	92.5	4.3	0.118
TFT, q=0.5	1.027	1.007	**0.996**	1.006	92.7	5.5	0.097
Seasonal naive, q=0.7	0.984	0.965	0.959	0.957	87.7	2.4	0.075
LightGBM, q=0.7	0.995	0.972	0.965	0.964	88.7	1.6	0.082
NHITS, q=0.7	1.011	0.985	0.971	0.976	89.1	1.1	0.101
TFT, q=0.7	1.010	0.985	0.974	0.977	89.4	1.6	0.085
Seasonal naive, q=0.9	0.930	0.910	0.904	0.897	79.6	0.6	0.070
LightGBM, q=0.9	0.914	0.892	0.887	0.879	78.2	0.5	0.071
NHITS, q=0.9	0.950	0.919	0.904	0.904	79.6	0.2	0.087
TFT, q=0.9	0.945	0.916	0.904	0.905	79.6	0.3	0.074

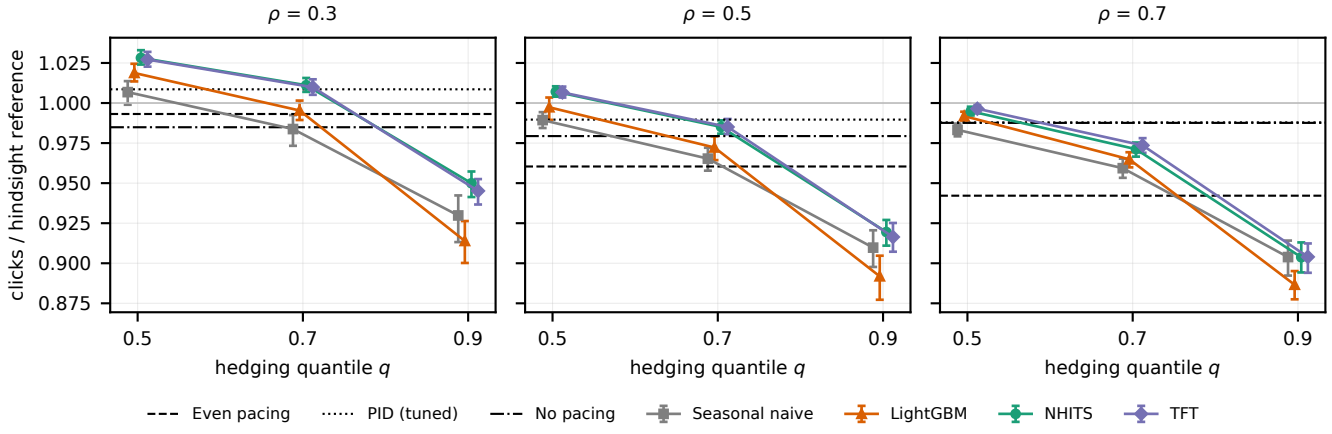


Figure 1: Click ratio versus planning quantile (95% campaign-bootstrap CIs). Horizontal lines are forecast-free policies.

Table 2: Original four forecasters: mean test accuracy (± training-seed SD). MASE_h is MASE at step h; Cov. is 80% prediction-interval coverage. Time includes fitting and forecasting.

Forecaster	MASE	$\text{MASE}_{h=1}$	$\text{MASE}_{h=24}$	$\text{sPL}_{0.9}$	Cov. 80%	Time (s)
Seasonal naive	0.917	0.920	0.908	0.194	60.6	0
LightGBM	0.853	0.810	0.871	0.185	71.7	76
NHITS	0.794 ± 0.014	0.719	0.813	0.158	75.8	40
TFT	0.787 ± 0.004	0.729	0.814	0.161	76.9	205

5 Experimental Protocol

Data and splits. The Criteo dataset contains 16,468,027 impressions from 675 campaigns over 742 hours. We use the first 30 complete days and discard the last 22 hours. Training uses 21 days, validation the next 4, and testing the last 5. Day boundaries are relative to the dataset start, not local midnight. The dataset license is CC BY-NC-SA 4.0.

We retain campaigns with nonzero spend in at least 90% of training hours. This gives 414 campaigns, representing 91.7% of dataset spend, and 2,070 test campaign-days. Each episode has 10 replay seeds. The selection rule uses training hours, but its threshold was chosen after a descriptive profile of the whole dataset, including test hours.

Forecast metrics. Mean absolute scaled error (MASE) [11] divides forecast error by the same campaign's training error from

a lag-24 forecast. A value below one beats that training scale; it does not necessarily beat the seasonal-naive model on the test set. We also measure pinball loss [13], coverage of the nominal 80% prediction interval, and quantile crossings. Scaled pinball loss (sPL) divides pinball loss by mean training hourly spend. Reported fitting and forecasting times use one RTX 3070 Ti for neural models and an Apple M4 CPU for LightGBM; they are not hardware-matched speed comparisons.

Pacing metrics. The primary metric is total clicks divided by the hindsight constant-rate reference's total clicks, pooled over episodes and seeds. A gain of one point means one percentage point of this ratio, or 1% of the reference click count. It is not a relative percentage gain over the compared policy. We also report attributed-conversion ratios, budget utilization (BUR, spend divided by budget), early depletion before hour 20, and the pacing volatility index (PVI, the root-mean-square hourly change in α_h). Spend changes and changes in clicks per unit of spend are relative percentages.

Uncertainty. Confidence intervals (CIs) use 2,000 campaign-bootstrap resamples, with paired differences between policies. The original 126 paired comparisons use Wilcoxon signed-rank tests [19] with Holm correction [10]. These tests use the 409 campaigns with positive reference clicks; other Criteo analyses use all 414. Training seeds are pooled within campaigns. Thus the intervals and tests condition on the saved trained models and do not include training-seed uncertainty. We report seed variation separately. We did not compute confidence intervals for MASE differences.

Protocol and deviations. A written protocol preceded the main evaluation, but test data were not entirely unseen. Besides the full-dataset profile, smoke tests inspected test-origin accuracy for seasonal naive, LightGBM and a short TFT run, and replayed ten campaigns' test days. The log reports no hyperparameter change in response to those test results. The PID search grid was widened using validation results. Failed GPU runs wrote no forecasts and were retried. At the final neural forecast origin, the preceding forecast was shifted by one hour because a full future window was unavailable.

The decomposition, scaling, hedging calibration, feedback, oracle and subgroup analyses were not planned before the main results were seen. New simulations were logged before execution, with decision rules where specified. The spend and subgroup analyses only reread saved episodes and were logged afterward. The original matched-spend comparison had no decision rule. Entry 21 later set rules before training the three additional neural models; this does not make the overall research sequence a single pre-registered test. The corrected dual-gradient variant was also added after its first results were seen.

Two protocol metrics are not reported here: hourly spend deviation from the reference profile, retained in `pacing_summary.csv`, and validation-split forecast metrics. The protocol originally specified a seven-day mean for seasonal naive; the implementation used empirical quantiles, hence the median at $q = 0.5$. An independent rerun rebuilt the data, retuned PID, regenerated seasonal-naive and LightGBM forecasts, and reproduced the pacing summary. It reused the neural forecasts with matching checksums; neural training was not independently repeated.

Table 3: Five-model extension. MASE averages campaigns and training seeds. Click gains and distances above the spend-matched naive curve are in reference-click points.

		Gain over seasonal naive			Above the scaling curve		
Model	MASE	ρ=0.3	ρ=0.5	ρ=0.7	ρ=0.3	ρ=0.5	ρ=0.7
TFT	0.787	2.03	1.73	1.32	1.20	0.74	0.47
NHITS	0.794	2.14	1.77	1.16	1.30	0.83	0.44
PatchTST	0.806	1.82	1.68	1.26	1.00	0.67	0.39
LSTM	0.816	1.60	1.27	0.98	0.81	0.43	0.26
TiDE	0.786	2.05	1.92	1.50	1.21	0.82	0.50

6 Results

6.1 Forecast accuracy

In the original comparison, mean MASE is lowest for TFT (0.787), followed by NHITS (0.794), LightGBM (0.853) and seasonal naive (0.917); see Table 2. All four are below the in-sample lag-24 scale. The later five-model comparison is shown in Table 3.

The mean ranking does not hold for every campaign. Median campaign MASE is 0.708 for TFT, 0.723 for NHITS and 0.722 for LightGBM. TFT has lower error than LightGBM in 70.5% of campaigns, and NHITS in 56.0%. These differences are descriptive, without paired accuracy CIs. Separately, the prediction intervals cover too few outcomes: the nominal 80% interval covers 76.9% for TFT and 60.6% for seasonal naive. Prediction-interval coverage and uncertainty about average MASE differences are different questions.

6.2 Pacing outcomes

Median-plan TFT and NHITS capture 3.4–5.4 points more clicks than even pacing and 0.7–2.0 points more than tuned PID (Table 1). Seasonal naive already adds 1.4–4.1 points over even pacing; the two neural models add 1.2–2.1 points beyond that. All 12 of 12 paired neural-versus-even-or-PID comparisons are significant after Holm correction (largest adjusted $p = 0.00052$). These tests condition on the trained models and do not include the later PatchTST, LSTM and TiDE runs.

TFT and NHITS also change their participation rates less sharply than PID. Their PVI is 0.08–0.12, versus 0.43–0.60 for PID. Their early-depletion rate is 3.4–6.1%, versus 12.8–31.9%. These are replay measurements, not evidence of stability in a live auction system.

TFT and NHITS are close, not proven equivalent. Their median-plan click ratios differ by at most 0.2 points. Across training seeds, the SD reaches 0.45 points for NHITS and 0.18 for TFT. Only 1 of 3 paired comparisons is significant. We have neither seed-aware inference nor an equivalence test, so we do not claim identical performance. Their gains over LightGBM span 0.3–1.0 points; the smallest gain is below NHITS's seed SD.

In an exploratory split by training-period campaign size and volatility, TFT's gain over seasonal naive is positive in all 18 group-and-budget cells (1.0 to 3.1 points). Gains are larger in the most volatile third at the two tighter budgets, but the intervals overlap and are not corrected for multiple comparisons.

Table 4: Median-plan comparisons. Click gains are reference-click points with 95% campaign-bootstrap CIs; spend and efficiency columns are relative percentage changes.

Comparison	ρ	Clicks (points)	Spend (%)	Clicks per spend (%)
TFT − seasonal naive	0.3	2.03 [1.57, 2.62]	1.9	0.15
	0.5	1.73 [1.36, 2.11]	1.8	−0.03
	0.7	1.32 [1.06, 1.63]	1.4	−0.03
Scaled naive − seasonal naive	0.3	0.73 [0.55, 0.93]	1.4	−0.69
	0.5	1.03 [0.82, 1.25]	1.9	−0.83
	0.7	1.24 [0.96, 1.53]	2.4	−1.12
TFT − scaled naive	0.3	1.30 [0.93, 1.77]	0.4	0.84
	0.5	0.69 [0.39, 1.03]	−0.1	0.81
	0.7	0.08 [−0.19, 0.37]	−1.0	1.11

All five tested neural models beat seasonal naive on Criteo. Table 3 adds PatchTST, LSTM and TiDE to the comparison. Seasonal-naive MASE is 0.917, and LightGBM MASE is 0.853. All 5 of 5 neural models gain 0.98–2.14 click-ratio points over seasonal naive across the three budgets, with every gain CI above zero. All 5 also lie above the spend-matched scaling curve described below, by 0.26–1.30 points.

Entry 21 required at least four of five models to show each finding at two of three budgets. Both findings meet that rule, whose label is "a property of the class." Here the label describes the tested architectures on these Criteo logs, not all neural predictors or other datasets. The matched-spend method itself originated as an exploratory analysis.

The models spend 1.2–2.1% more than seasonal naive. Changes in clicks per unit of spend range from −0.15% to 0.24%; these point estimates are small but do not establish equivalence in efficiency. Ranking the five models by mean MASE gives the reverse of their ranking by mean click gain (Spearman correlation −1.00). This is descriptive: five models are too few to establish a general relationship, and TFT and NHITS have nearly equal mean gains. PatchTST remains competitive despite using no extra covariates.

6.3 Where the gain comes from

More spend, with little change in efficiency. TFT spends 1.4–1.9% more than seasonal naive (Table 4). Clicks per unit of spend change by −0.03% to 0.15%, with CIs including zero at 3 of three budgets. NHITS shows a similar pattern. Thus the extra clicks over this baseline mainly accompany more budget spent, rather than a measured improvement in clicks per unit of spend.

The comparison with PID differs. TFT's relative spend change is −0.3% to −0.1%, while clicks per unit of spend rise by 1.1–2.1%. The efficiency CIs are above zero at all three budgets. This describes where the click gain appears; it does not identify a causal auction-selection mechanism.

Scaling the naive forecast. Lowering the forecast denominator makes the throttle keep more impressions. We select scales of 0.85, 0.825 and 0.775 on validation days for the tight, medium and loose budgets. None is at the grid edge. On test days, this captures 36%, 60% and 94% of TFT's gain over unscaled seasonal naive. The scaled plan also spends more, but its clicks per unit of spend fall (Table 4).

TFT retains a positive gain over the scaled plan at the tight and medium budgets. At the loose budget, the difference is 0.08 points with CI [−0.19, 0.37], so there is no clear separation. By entry 19's rule, "an edge beyond one scale remains"; scaling does not explain almost all of the gain at every budget. This is one global scale per budget level, not a separate correction for each campaign or day.

Swapping the daily total and hourly profile. To separate these two parts of the forecast, we build fixed plans from the forecasts issued at the start of the day. We replace either the naive plan's daily spend total or its hourly shares with TFT's, leaving the other component unchanged (Table 5). These are spend totals and spend shares, not traffic counts or impression-arrival profiles.

Replacing only the total captures 68–82% of the gain from replacing both components. Replacing only the profile adds less, but its CI is above zero at the medium and loose budgets. The protocol therefore attributes the gain to the total at the tight budget and reports a partial result at the other two. The two component gains need not sum to the combined gain. Updating forecasts hourly adds a further 0.25–0.51 points for TFT relative to its fixed day-start plan. This experiment was run for TFT, not all five neural models.

Perfect knowledge under the same rule. Replacing TFT's forecasts with actual remaining logged spend demand adds 1.03–1.54 points (Table 5). The protocol calls this moderate headroom. The oracle is an information benchmark for this particular throttle, not a proven upper bound on clicks or on what another pacing rule could achieve. Its information still concerns the stream selected by the original logging policy.

Matching aggregate spend (exploratory). We use the 25 naive scales to trace a curve of total spend against total clicks. For each model, we linearly interpolate this curve at that model's aggregate spend. The five neural models lie 0.26–1.30 click-ratio points above it, with every CI above zero (Table 3). LightGBM is also above the curve at the tight budget (0.45 points, CI [0.13, 0.77]); its medium- and loose-budget intervals include zero.

The comparison matches overall spend, not spend within every campaign-day. It shows that one global rescaling does not reproduce the neural outcomes. It cannot distinguish campaign-specific total predictions, hourly allocation and hourly re-planning, or establish that the models select intrinsically better auctions. The analysis was introduced after the scaling results and had no initial decision rule. Its intervals also exclude training-seed uncertainty.

Another reactive baseline. The flat-target dual-gradient rule has a click-ratio difference of −4.8 to −3.0 points relative to tuned PID. TFT and scaled naive both beat it, meeting entry 20's rule that "the forecast advantage survives a tuned dual-gradient pacer." A profile-target variant under-spends. After seeing that result, we normalized its hourly error by $B/24$ instead of the hour-specific target. TFT still leads the corrected variant by 2.9–3.6 points. PID remains the strongest reactive rule tested here; this does not rule out better controller designs or tuning.

6.4 Hedging

Higher hourly quantiles leave more budget unspent. Across the four original forecasters, clicks fall as q rises from 0.5 to 0.7 to 0.9 in all 12 of 12 forecaster-and-budget combinations. Moving from 0.5 to 0.9 costs 7.7–10.6 click-ratio points. All 24 comparisons of 0.9 plans with even pacing and PID favor those baselines; 24 are significant after the original Holm correction.

Table 5: Click-ratio gains in points; brackets give 95% campaign-bootstrap CIs. Swaps hold day-start plans fixed; the oracle updates remaining spend each hour.

Comparison	Tight ($\rho = 0.3$)	Medium ($\rho = 0.5$)	Loose ($\rho = 0.7$)
Fixed plan minus fixed seasonal-naive plan			
TFT total, TFT profile	1.52	1.35	1.07
TFT total, naive profile	1.25 [0.82, 1.83]	1.02 [0.71, 1.38]	0.72 [0.50, 0.95]
Naive total, TFT profile	0.16 [−0.02, 0.31]	0.17 [0.05, 0.29]	0.25 [0.14, 0.37]
Oracle remaining spend − TFT	1.03 [0.73, 1.35]	1.50 [1.23, 1.78]	1.54 [1.27, 1.81]

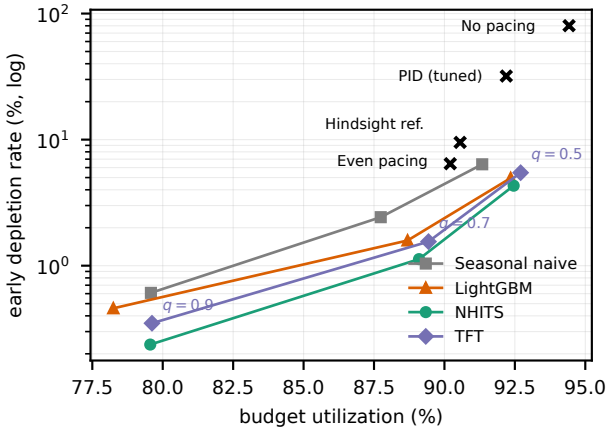


Figure 2: Budget utilization versus early depletion at $\rho = 0.5$. The depletion axis is logarithmic; lines connect $q = 0.5, 0.7, 0.9$.

The upper plans reduce early depletion to 0.1–0.7% of episodes, but budget utilization falls to 75.2–82.8% (Figure 2). For TFT, the relative spend change versus even pacing is −7.4% to −5.7%, despite an increase in clicks per unit of spend. Fewer clicks accompany less budget spent.

The objective here rewards clicks, without a separate penalty for exhausting the budget early. At the loose budget, no pacing and tuned PID have nearly equal click ratios (0.988 and 0.988). Thus the observed cost of hedging partly reflects this environment. It is not evidence that early exhaustion is unimportant to advertisers.

Hourly coverage does not determine total coverage. A sum of hourly quantiles is not generally the corresponding quantile of the remaining-day total. We measure this difference rather than assume a particular dependence between hourly errors. TFT's hourly 90th-percentile forecasts cover 90.8% of outcomes. Across the four forecasters, the summed 90% plans cover 94.0–97.4% of actual remaining spend totals at the evaluated origins, more than the nominal 90%. The summed plans are 1.80–2.06 times their median plans. A diagnostic fitted to those test outcomes gives smaller factors, 1.62–1.70, for 90% coverage. These diagnostic factors are not a guarantee of future coverage.

Calibration recovers part of the loss. We next estimate factors using earlier test days only and evaluate on the remaining 4 days. At $q = 0.9$, TFT recovers 20%, 14% and 11% of its summed-plan loss at the tight, medium and loose budgets. Each improvement CI is above zero, but the recovery falls below the protocol's one-third threshold. The rule therefore gives a partial verdict, not that calibration fails entirely or explains most of the cost.

Calibration at $q = 0.7$ recovers more of the loss, but TFT's calibrated-minus-median differences remain negative: −0.23, −0.87 and −1.37 points. For seasonal naive, calibrated 0.9 plans are worse than summed plans. The cross-fitted variant is no better for TFT at 0.9 and is not an upper bound on clicks. These results do not establish how much loss a different calibration method could recover.

The objective also matters. No pacing captures 1.4–3.2 points more attributed conversions than the hindsight reference while capturing fewer clicks. The policy ordering for clicks need not be the ordering for attributed conversions; neither measures causal conversion lift.

6.5 Adding feedback

Feedback on the day-start plan helps TFT at the tight budget but lowers its click ratio at the other two (Table 6). Tracking the previous hour's updated plan does not give a clear TFT improvement at either tighter budget and lowers clicks at the loose one. Entry 18 required a positive feedback-gain CI at two of three budgets. Neither variant meets that rule: the verdict is "feedback does not help TFT."

The same rule tests whether TFT with feedback beats scaled naive with feedback. It does not do so at two of three budgets in either experiment, giving "the neural edge does not survive feedback." This is a verdict about these feedback designs, not about all combinations of forecasting and control. The day-start experiment uses the earlier scale-grid baseline; the hourly-plan experiment uses the extended-grid baseline. They are not identical policy comparisons.

The gains were tuned on seasonal naive, not on the neural forecasts. Some selected gains reach the grid edge, and tuning uses only 4 validation days. At the loose budget, seasonal-naive feedback gains 1.0 points on validation days but changes test clicks by −0.82 points (CI [−2.50, 0.17]). Gain transfer and short-window tuning limit the interpretation. We did not establish why feedback reduces TFT's clicks, so we do not attribute the loss to jitter or overreaction.

6.6 A second environment: a synthetic audience world

We also use a generated opportunity stream, where the logging policy and cost process are under our control. The generator creates 50,000 users, 40 campaigns and 28 days, with planted daily and weekly patterns and user churn (mean lifetime 14 days). The pacers know the generated click probabilities, removing click-estimation

Table 6: Feedback comparisons in click-ratio points (95% campaign-bootstrap CIs). Each block uses its own validation-tuned gains and scaled-naive baseline.

Comparison	Tight ($\rho = 0.3$)	Medium ($\rho = 0.5$)	Loose ($\rho = 0.7$)
Feedback tracks the day-start plan			
TFT with − without feedback	0.50 [0.23, 0.79]	−0.97 [−1.68, −0.41]	−2.01 [−2.79, −1.43]
TFT − scaled naive, both with feedback	0.26 [−0.02, 0.59]	0.38 [−0.31, 1.42]	−1.92 [−2.75, −1.26]
Feedback tracks the previous hour's plan			
TFT with − without feedback	0.14 [−0.02, 0.29]	−0.12 [−0.34, 0.10]	−1.64 [−2.04, −1.29]
TFT − scaled naive, both with feedback	−0.19 [−0.43, 0.05]	0.27 [0.10, 0.46]	−1.39 [−1.75, −1.03]

error from this comparison. This is a controlled pacing experiment, not a validated model of a live advertising market.

Across 6 worlds and 4 cost models, replaying the previous seven days' opportunities gains 1.5–2.7 points over reactive control. Here points are shares of a Lagrangian hindsight value bound, not the Criteo click reference. Rescaling the replay to the true expected traffic total adds only 0.1–0.3 points; all intervals for that addition include zero. The combined gain over reactive control is 1.6–3.0 points, with a CI above zero only for time-of-day pricing.

Historical replay therefore captures nearly all of the observed forecasting benefit in these worlds. This leaves little measured room for a more accurate forecaster in that setting, rather than confirming a neural advantage. The rhythms were built into the generator and daily totals vary less than in the real logs. We do not use this result to establish cross-dataset generalization of the Criteo findings.

6.7 A second real dataset: iPinYou

The iPinYou season-2 logs provide 12.2 million won impressions from 5 advertisers over seven days. We treat each advertiser as a campaign: the first four days are training, one is validation, and two are test days, giving 10 test episodes. Forecasts use the same hour on the previous three days. We do not train or test neural predictors on this dataset. Intervals resample test episodes rather than campaigns.

Entry 20 calls the replication consistent if scaled naive ranks at least as high as naive, and naive at least as high as even pacing and PID, at two of three budgets. This ordering holds at 0 levels, so the descriptive verdict is "inconsistent." Naive-minus-even click differences are −1.1, −1.1 and −0.8 points, with every CI including zero. No pacing has the highest realistic-policy point estimate at the tight budget.

The oracle remaining-spend plan beats naive by 6.4 points (CI [1.6, 15.1]), 5.5 ([1.8, 10.8]) and 4.9 ([1.3, 10.5]). This shows an advantage for perfect logged-spend information under the same rule. It does not show that a feasible learned model can obtain it. With few advertisers, two test days, one tuning day and a selected scale at the grid edge for the loose budget, the comparison neither reproduces the Criteo ordering nor precisely establishes a contrary ordering.

7 Discussion

What the evidence supports. All five tested neural models improve clicks over seasonal naive on Criteo. Their extra clicks accompany more spend and small changes in clicks per unit of spend. TFT's swap experiment points mainly to its daily spend total rather than its hourly profile. A single validation-chosen scale captures much of TFT's gain at the loose budget, but less than half at the tight one. It is therefore useful to compare a neural model with a scaled seasonal baseline, not only an unscaled one.

The remaining gain at matched aggregate spend is a result to explain, not a mechanism already established. The comparison does not tell us whether it comes from campaign-specific totals, hourly profiles or re-planning. Nor does the five-model ranking establish that lowering MASE will reliably raise clicks in a different setting. NHITS and TFT have close pacing outcomes, while their recorded fitting-and-forecasting times differ (40 s and 205 s on the same GPU setup). That is a practical compute difference in these runs, not proof that the architectures are interchangeable.

Implications for this replay setting. The median is the best of the three planning quantiles tested for the original four forecasters. Lower quantiles were not evaluated for pacing, so we do not claim that the median is optimal. Better coverage of the remaining-spend total also does not automatically give more clicks: a conservative plan can still leave budget unspent. A system with explicit delivery or early-exhaustion penalties may make a different trade-off.

Feedback should likewise be tested with the forecast and objective that will actually be used. Transferring gains from seasonal naive did not consistently improve TFT. This does not justify a blanket rule to add feedback only at tight budgets. The small iPinYou comparison and the generated worlds also do not establish where neural forecasting will work outside Criteo.

Limits of the evidence. The main evaluation spans 5 days of one dataset. Campaign bootstrap intervals describe variation across campaigns on those days, not uncertainty from a new test period or retraining. All tests condition on saved neural forecasts; LightGBM and seasonal naive have single runs. There are no paired accuracy CIs or seed-aware pacing tests. In Table 4, CIs are shown only for clicks; the TFT-versus-scaled-naive efficiency difference has no CI.

The replay uses only won impressions and transformed prices. It omits competitive responses, price effects and a separate cost of early exhaustion. The reference is not a ceiling and the oracle knows logged spend, not the full auction market. Day boundaries are relative to the dataset start. These restrictions limit both economic interpretation and deployment advice.

The main design was informed by a full-data profile and test smoke checks. Later experiments reused the same test period after the main results were known. Follow-up CIs are not a new Holm-corrected family of tests, and the subgroup CIs are not corrected for multiple comparisons. Gains and scales were tuned on only 4

validation days, with some controller settings at search boundaries. The high PID volatility may indicate scope for a better controller, but does not diagnose why feedback loses clicks. More campaigns do not remove these time-period, tuning and design limits.

8 Conclusion

On Criteo, all five tested neural predictors capture more clicks than seasonal naive under the same throttling rule. Most of TFT's improvement in the day-start swap experiment comes from its daily spend total. Scaling the seasonal baseline recovers 36–94% of TFT's gain, depending on budget tightness. A smaller gain remains at matched aggregate spend, but its cause is unresolved.

Upper-quantile plans leave more budget unspent in this click-focused replay. Calibrating the total recovers part of the loss, and the tested feedback designs do not consistently improve TFT. These findings are limited by a short test period, fixed-model inference and the logging-policy assumptions. The iPinYou study does not reproduce the policy ordering and tests no neural models. The next empirical steps are seed-aware inference, longer out-of-time evaluations on another real dataset, and a simulator that tests price responses and explicit delivery constraints.

Use of Generative AI Tools

Claude (Anthropic) and an OpenAI assistant helped with code, experiments, writing and draft review. Numerical results come from the accompanying code. Model feedback is not human peer review. The author is responsible for the content.

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